

CSIT 595 Special Topics for GenAI

Assignment 1

Due: 23:59 Oct 5, 2026

Objective

In this assignment, you will implement and evaluate **three generative models** on the **same dataset**:

- Variational Autoencoder (VAE)
- Generative Adversarial Network (GAN)
- Diffusion Model (DDPM-style)

Your goal is to compare these models through a practical **generative trilemma**:

1. Sample quality (realism)
2. Diversity / coverage (mode coverage)
3. Sampling cost (speed at inference)

Note: This is an **unconditional** assignment. You may **NOT** use labels for conditioning (no class embeddings, no classifier guidance, no prompts).

Dataset

Use **Fashion-MNIST**

- Images: 28×28 grayscale.
- For consistency across models, convert inputs to $32 \times 32 \times 1$ (pad or resize).
- Normalize pixel values to $[-1, 1]$.

Tasks

Common Setup + Evaluator Network (3 points)

You will build a shared evaluation pipeline used for **all three** generative models.

- **Preprocessing:** Implement a dataloader with the required format ($32 \times 32 \times 1$, normalized to $[-1, 1]$).
- **Evaluator CNN:** Train a small CNN classifier on Fashion-MNIST. Report test accuracy.
- **Feature extractor:** Use the penultimate-layer features from the evaluator CNN to compute a lightweight “FID-like” score, called **cFID**.
- **Coverage metrics:** Using the evaluator predicted labels on generated images, compute:
 - Generated label histogram (10 classes)
 - Label entropy
 - KL divergence between generated label distribution and real label distribution
- **Sampling cost:** Measure sampling speed (images/sec) and report time to generate 5,000 images.

1. Variational Autoencoder (VAE) (4 points)

Implement an unconditional convolutional VAE.

- Train the VAE to minimize reconstruction loss + KL divergence.
- Report training curves:
 - Reconstruction loss
 - KL divergence term
 - Total loss
- Required visualizations (include in your report):
 - Reconstruction grid (original vs reconstruction)
 - 100 unconditional samples from prior $z \sim \mathcal{N}(0, I)$
 - Latent interpolation between two images (at least 8 interpolation steps)
- Evaluate and report:
 - cFID, coverage metrics, sampling speed

2. GAN (4 points)

Implement an unconditional GAN. DCGAN-style is recommended, but WGAN-GP is allowed if you prefer stability.

- Report training curves:
 - Generator loss
 - Discriminator loss
- Required visualizations (include in your report):
 - 100 unconditional samples
 - Latent interpolation (at least 8 interpolation steps)
 - **Mode collapse watch:** plot the evaluator-predicted label histogram of generated samples at multiple training checkpoints
- Evaluate and report:
 - cFID, coverage metrics, sampling speed

3. Diffusion Model (6 points)

Implement an unconditional diffusion model that predicts noise $\epsilon_\theta(x_t, t)$ using a small U-Net.

- **Forward process:** Define a noise schedule with T timesteps (e.g., $T = 500$).
- **Training:** Train by sampling a random t each update and minimizing MSE between true noise ϵ and predicted noise $\hat{\epsilon}$.
- Report training curve: noise-prediction MSE vs training steps.
- **Sampling:** Implement sampling and report results at **multiple step counts** to show the speed-quality tradeoff.
 - For each step count:
 - * 100 unconditional samples (grid)
 - * cFID and coverage metrics
 - * sampling speed (images/sec)

4. Performance Comparison and Trilemma Analysis (3 points)

Summarize and compare VAE vs GAN vs Diffusion.

- Create a single results table including :
 - cFID
 - Coverage: entropy and KL divergence
 - Sampling speed: images/sec and time to generate 5,000 images
 - Number of parameters and total training time
- Write a short discussion answering:
 - Which model performs best on **quality**? on **diversity/coverage**? on **sampling speed**?
 - What failure modes did you observe? (e.g., VAE blur, GAN instability/mode collapse, diffusion slow sampling)
 - For diffusion: how do fewer sampling steps change quality vs speed?

Training Budget Constraint

Your implementation should be designed so that training for **all three models** can finish within **one day** on Google Colab Pro.

- If you hit time/compute limits, reduce model size and/or training steps.
- You must report your final training budget (epochs/steps, batch size) and runtime for each model.

Submission Guidelines

- Assignment should be submitted via Canvas.
- Submit a single `.zip` file that contains:
 - Codes in `.ipynb` format in one folder named *codes*.
 - A report (Word or $\text{\LaTeX}$ /PDF) summarizing:
 - * Architectures + hyperparameters
 - * Training curves and runtime
 - * Sample grids + interpolations
 - * Metrics table + trilemma analysis